



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn; A Chance to Grow"
2025-26 (Even Sem)



BT230004CS

MSE 2 EXAMINATION

Course: B. Tech. In CSE

Subject: Elective-4 (No SQL & New SQL)

Max Marks: 20

Date: 04/04/2026

Year/Semester: 3rd Year / 6th Semester

Subject Code: CS6E004A

Duration: - 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Understand the characteristics and need for NoSQL and NewSQL systems.
2. Model, store, and query data using key-value, document, column-family, and graph databases.
3. Critically evaluate consistency, availability, and partition tolerance in NoSQL and NewSQL systems.
4. Assess performance, fault tolerance, replication, and sharding techniques in distributed databases.
5. Design and implement end-to-end mini-projects or case studies using NoSQL/NewSQL databases for applications in domains.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define the term Keyspaces.	L1/CO2	1
Q.2	List out the datatypes in Cassandra Query Language.	L1/CO3	1
Q.3	Define node and relationship in graph databases.	L1/CO2	1
Q.4	State the advantages of Column-Family stores.	L1/CO2	1

SECTION B

1. All Questions are compulsory.
2. Answer in two or three sentences.

Q.5	Differentiate between Cassandra column family and relational database.	L4/CO2	2
Q.6	Explain data modeling in Cassandra	L2/CO2	2
Q.7	Describe the features of Cassandra.	L2/CO2	2
Q.8	Explain how Neo4j differs from relational databases.	L2/CO3	2

SECTION C

1. Solve any two questions.

Q.9	Describe the structure of a graph database.	L2/CO2	4
Q.10	Explain the architecture of Apache Cassandra.	L2/CO2	4
Q.11	Examine the replication strategies in Cassandra.	L3/CO3	4